

ANDREW AMAVIZCA

SOFTWARE & ROBOTICS ENGINEER | AUTONOMOUS SYSTEMS & PERCEPTION

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EDUCATION

University of California, Merced — B.S. Mechanical Engineering, May 2025

Focus: Controls, Embedded Systems, Systems Integration, Data-Science

CORE SKILLS

Programming: Python, C++, Embedded C/C++, MATLAB, Object-Oriented Programming (OOP)

Software & Tools: Git, Linux, Jira, simulation, automation tooling, software regression testing

Autonomy & Robotics: Perception pipelines, sensor integration, autonomous sensing systems, health monitoring, anomaly detection, V&V

Sensors & Hardware: LiDAR, cameras, remote sensing, embedded sensors, IoT devices, unmanned systems (UGV/UAV), modular sensor platforms

Machine Learning & Data: Perception-focused ML, explainable AI (XAI), large-scale sensor data analysis

Systems Engineering: Systems integration, requirements-driven design, field testing, hardware/software integration

EXPERIENCE

AUTONOMOUS VEHICLE TEST ENGINEER L1 — TRANSDEV ONSITE W/ WAYMO, ATWATER, CA | APRIL 2026—PRESENT

- Performed V&V analysis of LiDAR and camera sensor data, evaluating sensor outputs and perception-system behavior across autonomous vehicle test scenarios.
- Conducted firmware regression testing and triage, comparing new and baseline releases in simulation to identify behavioral changes, software regressions, and anomalous system responses.
- Investigated test failures by analyzing system outputs and sensor data, isolating unexpected behavior and documenting findings for further software debugging and root-cause analysis.
- Developed and executed test plans for sensing and perception software, defining test conditions, expected system behavior, acceptance criteria, and validation requirements.
- Documented and tracked software defects, validation results, and reproducible issues in **Jira**, supporting engineering triage and subsequent software releases.

ASSOCIATE MANUFACTURING ENGINEER — APPLIED AEROSPACE, STOCKTON, CA | OCT 2025—APRIL 2026

- Developed Python automation tools that eliminated repetitive engineering work, recovering ~5 engineer-hours/day per engineer and saving tens of thousands annually.
- Interpreted engineering drawings, Conditions of Supply, and EO/NIEO data to define system interfaces, tolerances, and manufacturability constraints for complex aerospace assemblies.
- Built complete BOMs and performed make-vs-buy analyses to support supplier quoting, cost decisions, schedule, and risk reduction.
- Authored test procedures, acceptance criteria, and temperature-controlled material-handling instructions for qualification-sensitive components.
- Worked cross-functionally with technicians, quality, and supply chain teams to resolve system-level hardware blockers and sustain production throughput under tight schedules.

SELECTED PROJECTS

AUTONOMOUS FLOOD IRRIGATION MONITORING SYSTEM — End-to-End System Owner

- Owned the full system stack from concept through deployment, spanning CAD design, component selection, electrical integration, embedded software, autonomous control logic, and field validation.
- Designed and built the mechanical and electrical architecture, including enclosures, sensor interfaces, power distribution, wiring, soldered connections, and embedded hardware.
- Developed embedded software and autonomous control logic for sensor acquisition, water-level monitoring, fault handling, and unattended system operation.
- Implemented system health monitoring for battery status, sensor state, signal integrity, and fault conditions to support reliable extended field deployment.
- Performed end-to-end debugging and validation across mechanical, electrical, and software interfaces, resolving integration issues and verifying operational readiness.

PERCEPTION & REMOTE SENSING PIPELINES (UAV/ SATELLITE) — Software & Perception

- Developed perception pipelines using machine learning to process and analyze noisy real-world sensor data
- Applied explainable AI techniques to improve model interpretability, validation, and failure analysis.
- Validated perception outputs through iterative testing with real-world data.

MODULAR UGV SENSOR PLATFORM — Systems Integration

- Designed modular sensor platforms for unmanned ground vehicles supporting scalable integration of heterogeneous sensors
- Defined standardized mechanical, electrical, and data interfaces to enable rapid sensor swaps and future expansion
- Collaborated with mechanical and software teams to validate integrated systems under operational conditions